[SYNTHETIC PIPELINE SANITY TEST — NOT REAL LITERATURE]

Mitochondrial Outer Membrane Permeabilization Releases Cytochrome C to Trigger Caspase Cleavage

Authors: Synthetic Benchmark Generator
Provenance: Synthetic Pipeline Sanity Test (Procedural Fixture)

Page 1 — Abstract & Introduction

Intrinsic apoptotic signaling is triggered by irremediable cell damage. Pro-apoptotic BAX oligomerizes in the mitochondrial outer membrane to induce permeabilization.

Page 2 — Mitochondrial Release

Permeabilization by BAX triggers cytochrome c release from the intermembrane space into the cytosol.

Page 3 — Apoptosome Formation

Cytosolic cytochrome c binds APAF-1, forming the apoptosome to stimulate Caspase-9 proteolytic cleavage and activation.

Page 4 — Executioner Cleavage

Active Caspase-9 directly cleaves and activates Caspase-3, driving substrate cleavage and nuclear condensation.

Page 5 — Discussion

We show that BAX permeabilization triggers Caspase-3 activation via intermediate Caspase-9 cleavage. Non-apoptotic controls did not cleave Caspase-3.